

Software Requirements Specification

for

Vehicle Rental Management System, Release 1.0

Version 1.0 approved

Prepared by Rohan Dhumal,

Nidhi Dhyani

Sardar Patel Institute of Technology, Andheri

July 28, 2026

Table of Contents

1. Introduction	3
1.1 Purpose	3
1.2 Project Scope and Product Features	3
1.3 References	3
2. Overall Description	3
2.1 Product Perspective	3
2.2 User Classes and Characteristics	4
2.3 Operating Environment	4
2.4 Design and Implementation Constraints	5
2.5 User Documentation	5
2.6 Assumptions and Dependencies	5
3. System Features	6
3.1 Book a Vehicle	6
3.2 Search Vehicles and View Availability	8
3.3 Take Delivery of and Return a Vehicle	8
3.4 Pay for a Rental and Receive Refunds	9
3.5 View Rental History	10
3.6 Create, View and Close Vehicle Maintenance Records	11
3.7 Create, View, Modify and Delete Fleet, Tariffs and Users	12
3.8 Notifications and Management Reports	13
4. External Interface Requirements	13
4.1 User Interfaces	13
4.2 Hardware Interfaces	14
4.3 Software Interfaces	14
4.4 Communications Interfaces	15
5. Other Nonfunctional Requirements	15
5.1 Performance Requirements	15
5.2 Safety Requirements	15
5.3 Security Requirements	16
5.4 Software Quality Attributes	16
5.5 Business Rules	17
Appendix A: Data Dictionary and Data Model	18
Appendix B: Analysis Models	19

Revision History

Name	Date	Reason For Changes	Version
Rohan Dhumal	28/07/25	initial draft	1.0 draft 1
Nidhi Dhyani	04/08/25	revised after internal review of ambiguities	1.0 draft 2
Rohan Dhumal, Nidhi Dhyani	04/08/25	baseline following changes after inspection	1.0 approved

1. Introduction

1.1 Purpose

This SRS describes the software functional and nonfunctional requirements for release 1.0 of the Vehicle Rental Management System (VRMS). This document is intended to be used by the members of the project team that will implement and verify the correct functioning of the system, and by the customer representative who will validate that it states the correct requirements. Unless otherwise noted, all requirements specified here are high priority and committed for release 1.0.

1.2 Project Scope and Product Features

The Vehicle Rental Management System will permit customers of <Agency Name> to book cars and two-wheelers on-line, to collect them from or have them delivered by any branch, to pay for them

electronically or in cash, and to view their complete rental history. It will also permit the agency to manage its fleet, its tariffs and the maintenance record of every vehicle. VRMS replaces the manual register and telephone process presently used, which causes double bookings, disputes over charges and vehicles that miss their scheduled service. A detailed project description is available in the Vehicle Rental Management System Vision and Scope Document [1].

The features scheduled for implementation in release 1.0 are:

- Vehicle search and real-time availability
- Booking creation, modification, extension and cancellation
- Vehicle handover and return, with photographic condition reports
- Payments, security deposits, invoicing and refunds
- Rental history and downloadable documents
- Vehicle maintenance records and preventive service scheduling
- Fleet, tariff and user administration
- Notifications and management reporting

The system is expected to evolve over several releases, ultimately connecting to insurance claim services, vehicle telematics and third-party travel marketplaces. Vehicles listed by private owners, chauffeur-driven bookings and loyalty programmes are outside the scope of release 1.0.

1.3 References

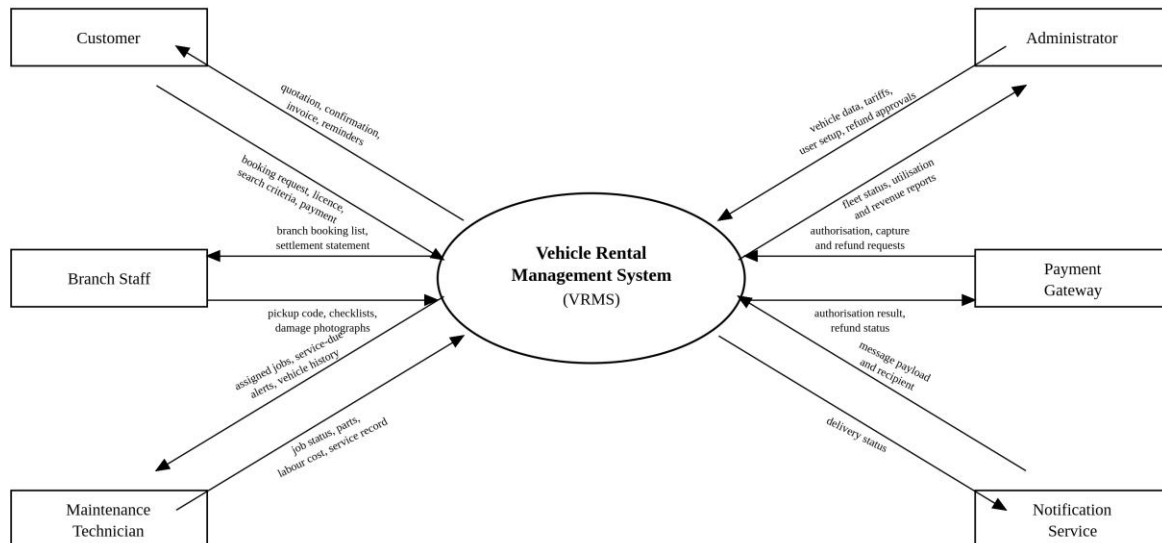
1. <Agency Name> Vehicle Rental Management System Vision and Scope Document, Version 1.0, <author>, <date>.
2. IEEE Std 830-1998, IEEE Recommended Practice for Software Requirements Specifications.
3. Wiegers, Karl. Software Requirements Specification for Cafeteria Ordering System, Release 1.0, Process Impact, 2002. (Format followed by this document.)
4. <Agency Name> Rental Terms, Conditions and Tariff Policy, Version <n>, <date>.
5. <Gateway Name> Payment Gateway REST API Reference, Version <n>, <URL>.
6. Web Content Accessibility Guidelines (WCAG) 2.1, W3C Recommendation, June 2018.

2. Overall Description

2.1 Product Perspective

The Vehicle Rental Management System is a new, self-contained system that replaces the current manual and telephone processes for reserving, issuing and returning vehicles at <Agency Name>. It is built as a responsive web application and an Android and iOS mobile application, both served by a single REST application server and one relational database. The context diagram in Figure 1 illustrates the external

VRMS depends on two external services. The Payment Gateway authorises, captures and refunds all electronic payments; VRMS itself never handles card credentials. The Notification Service delivers e-mail, SMS and push messages.



Customer	registration and licence details, search criteria, booking request, payment authorisation, rating → search results, price quotation, booking confirmation, invoice, reminders
Branch Staff	pickup code, handover and return checklists, damage photographs, cash receipt → branch booking list, settlement statement
Maintenance Technician	job status, parts and labour costs, service record → assigned jobs, service-due alerts, vehicle history
Administrator	vehicle master data, tariffs, user setup, refund approvals → fleet status, utilisation and revenue reports, document-expiry alerts
Payment Gateway	authorisation, capture and refund requests → authorisation result, settlement webhook, refund status
Notification Service	message payload and recipient → delivery status

Figure 1. Context diagram for release 1.0 of the Vehicle Rental Management System.

2.2 User Classes and Characteristics

Customer (favoured)	A member of the public who books a car or two-wheeler. An estimated 5,000 customers will register in the first year, of whom about 1,200 are expected to be active each month, placing an average of 200 bookings per day and up to 400 at weekends (source: current manual booking register). About 70 percent of bookings are expected from the mobile application and 30 percent from the web. Technical expertise varies widely, so no customer function may require training: customers must be able to book, modify, extend, cancel and pay without assistance from staff. This is the most important user class to satisfy.
Branch Staff	About four employees at each of five branches. They verify the identity of the customer, complete the handover and return checklists, record cash payments, settle charges and flag vehicles for maintenance. They use the system continuously during branch hours on a shared tablet and will need to be trained in its use.

Maintenance Technician	About six technicians, employed by the agency or working at a contracted workshop. They receive maintenance jobs, record the work performed, the parts consumed and the cost, and close the job. They use the system a few times a day on a mobile device and have low to moderate computer literacy, so their screens must be simple and tolerant of interruption.
Administrator	One to three users, responsible for vehicle master data, tariffs, branches, user accounts and refund approvals. The Administrator is the only user class permitted to change a tariff (BR-13) and is assumed to be technically competent.
Branch Manager	One per branch. A restricted Administrator whose authority and reporting visibility are limited to the vehicles, staff and bookings of the branch to which the user is assigned.
Guest	Any visitor who browses the catalogue and checks prices and availability without logging in and may not place a booking. This class is commercially important as the source of new customers, but is the least privileged.

2.3 Operating Environment

- OE-1: The Vehicle Rental Management System shall operate with the current and one previous major version of the Google Chrome, Mozilla Firefox, Microsoft Edge and Apple Safari web browsers.
- OE-2: The mobile application shall operate on Android 9.0 and later and on iOS 14 and later.
- OE-3: The web client shall be fully usable at screen widths from 360 pixels to 1920 pixels, so that the same application serves customers on a telephone and staff on a tablet or desktop.
- OE-4: The system shall operate on a server running Ubuntu Server 22.04 LTS or later, with the application server and the database deployed as containers on a host of 4 vCPU and 8 GB RAM.
- OE-5: The system shall use PostgreSQL 14 or later, or MySQL 8 or later, as its sole system of record.

2.4 Design and Implementation Constraints

- CO-1: All communication between clients, the server and every external service shall use HTTPS with TLS 1.2 or higher.
- CO-2: The system shall not store, log or transmit card numbers, expiry dates or CVV values. All card payments shall be handled by the payment gateway using hosted fields or tokens, so that the deployment qualifies for PCI DSS SAQ-A.
- CO-3: All business logic, in particular availability, pricing and authorisation checks, shall be enforced on the server. Client-side validation is provided for usability only and shall never be the sole control.
- CO-4: The system shall expose its functionality through a REST API using JSON, documented with OpenAPI 3.0, so that the web and mobile clients share one server.
- CO-5: The database schema shall be normalised to at least third normal form and referential integrity shall be enforced by database constraints.
- CO-6: All timestamps shall be stored in UTC and displayed in local time. All monetary values shall be stored in a fixed-point decimal type and never in a floating-point type.
- CO-7: Only third-party components under the MIT, Apache 2.0 or BSD licences shall be used.
- CO-8: Because the system must be delivered within one academic semester by a team of two developers, the payment gateway, e-mail and SMS services shall be integrated in sandbox mode. Live financial settlement is outside the delivery scope.

2.5 User Documentation

- UD-1: The system shall provide an online, cross-linked help system in HTML that describes and illustrates all system functions, with a help link on every screen.
- UD-2: The first time a new customer accesses the system, and on demand thereafter, the system shall provide an online tutorial that allows the customer to practise placing a booking using a static demonstration fleet. The system shall not store bookings made in the tutorial and shall not reserve any real vehicle.
- UD-3: A staff and technician operating manual shall be delivered covering handover, return, settlement, cash handling and maintenance job recording.
- UD-4: An administrator manual shall be delivered covering fleet setup, tariff definition, user management, reporting, backup and restore.

2.6 Assumptions and Dependencies

- AS-1: It is assumed that the agency owns every vehicle offered through the system, and that vehicles listed by private owners are not supported in release 1.0.
- AS-2: It is assumed that every customer holds a valid driving licence appropriate to the vehicle category and is willing to upload images of it for verification.

AS-3:	It is assumed that every branch has a reliable Internet connection during working hours and at least one camera-equipped tablet for condition reporting.
AS-4:	It is assumed that odometer and fuel readings are entered honestly by staff at handover and return. The system verifies internal consistency but cannot verify the physical reading.
DE-1:	The confirmation of an on-line booking depends on the availability of the external payment gateway. If the gateway is unreachable, the booking remains unconfirmed and no vehicle is committed.
DE-2:	Booking confirmations, one-time passwords and reminders depend on the external e-mail and SMS providers. Failure of these providers does not invalidate a booking but does degrade the requirements of Section 3.8.
DE-3:	The correctness of preventive maintenance scheduling depends on the odometer reading being captured at every return (Return.Odometer).

3. System Features

3.1 Book a Vehicle

3.1.1 Description and Priority

A customer whose identity and driving licence have been verified may reserve a specific car or two-wheeler for a specific period, either to be collected at a branch or to be delivered to an address. A customer may modify, extend or cancel a booking under the rules stated in Section 5.5. Priority = High.

3.1.2 Stimulus/Response Sequences

Stimulus:	Customer requests to book a selected vehicle.
Response:	System checks eligibility, places a temporary hold on the vehicle and queries the customer for add-ons, delivery instructions and payment.
Stimulus:	Customer requests to change a booking.
Response:	If status is “Confirmed” and the vehicle is free for the new period, system allows the customer to edit the booking and recalculates the price.
Stimulus:	Customer requests to cancel a booking.
Response:	If status is “Confirmed,” system states the refund due and cancels the booking.
Stimulus:	Customer requests to extend an active rental.
Response:	If no later booking conflicts, system quotes the additional charge and extends the return time.

3.1.3 Functional Requirements

Book.Create:	The system shall let a customer who is logged into the Vehicle Rental Management System place a booking for one vehicle for a specified pickup branch, pickup date and time, and return date and time.
Book.Eligible.Licence:	The system shall confirm that the customer's driving licence has been verified, that it authorises the category of the vehicle, and that it does not expire before the return date (see BR-2 and BR-3).
Book.Eligible.No:	If the licence has not been verified, the system shall give the customer the options to submit a licence now, to continue browsing without booking, or to exit.
Book.Period.Limits:	The system shall enforce a minimum rental period of 4 hours and a maximum of 30 days, counting any extension towards the maximum (see BR-5).
Book.Hold:	When the customer proceeds to payment, the system shall place a temporary hold on the vehicle for 15 minutes, during which the vehicle shall not be offered to any other customer.
Book.Hold.Expire:	If payment is not completed within the hold period, the system shall release the vehicle, shall inform the customer that the hold has expired, and shall retain the entered details so that the customer can try again.

Book.NoDouble:	The system shall never permit two bookings with a status of Confirmed or Active to overlap in time for the same vehicle. This check shall be performed at the moment of confirmation and not only at the moment of search.
Book.Concurrent:	If two customers attempt to confirm the same vehicle for overlapping periods at the same time, the system shall confirm exactly one booking and shall inform the other customer that the vehicle is no longer available, without charging that customer.
Book.Addons:	The system shall let the customer add an additional helmet, a child seat, a luggage carrier, an additional named driver or doorstep delivery to the booking, charged per BR-7.
Book.Deliver.Location :	If doorstep delivery is requested, the customer shall provide a delivery address, and the system shall confirm that it lies within the serviceable area of the branch.
Book.Quote.Display:	When the customer indicates that no further add-ons are required, the system shall display the base rental charge, each add-on charge, the taxes, any discount and the total payable, calculated per BR-6, together with the refundable security deposit shown as a separate amount.
Book.Quote.Terms:	The system shall display the free-kilometre allowance, the excess-kilometre rate, the fuel policy, the late return charge and the cancellation slabs, and shall require the customer to accept the rental terms before payment.
Book.Confirm.Prompt:	The system shall prompt the customer to confirm the booking.
Book.Confirm.Pay:	A booking shall be confirmed only after the payment authorisation succeeds (Pay.Authorize), or, for a booking to be collected at a branch, after a staff member records a cash payment (see BR-9).
Book.Done:	When the customer has confirmed the booking and payment has been authorised, the system shall do the following as a single transaction:
Book.Done.Store:	Assign the next available booking reference and a six-digit pickup code to the booking, and store it with an initial status of "Confirmed."
Book.Done.Block:	Mark the vehicle as unavailable for the booked period together with the turnaround buffer defined in BR-1.
Book.Done.Customer:	Send an e-mail message to the customer with the booking, price and collection details, and an SMS message with the pickup code.
Book.Done.Branch:	Add the booking to the collection schedule of the pickup branch and notify the branch staff.
Book.Done.Failure:	If any step of Book.Done fails, the system shall roll back the transaction, release the hold, reverse any payment authorisation obtained, and notify the customer that the booking was unsuccessful, along with the reason for failure.
Book.Modify:	The system shall permit the customer to change the period, the pickup branch or the add-ons of a Confirmed booking, provided that the request is made at least 12 hours before the pickup time and the vehicle is free for the new period.
Book.Modify.Price:	On modification the system shall recalculate the price, charge any additional amount immediately, and refund any excess under the cancellation rules of BR-8.
Book.Cancel:	The system shall permit the customer to cancel a Confirmed booking at any time before the vehicle is handed over. The system shall display the refund computed per BR-8 and shall require confirmation before cancelling.
Book.Cancel.Release:	On cancellation the system shall immediately release the vehicle, set the booking status to "Cancelled," initiate the refund and notify the customer and the branch.
Book.Extend:	The system shall permit the customer to extend an Active rental only if no confirmed booking of the same vehicle conflicts with the extended period and the total period remains within BR-5. The additional charge shall be collected before the extension takes effect.

Book.Status: The system shall maintain every booking in exactly one of the statuses Draft, Pending Payment, Confirmed, Active, Completed, Cancelled or No-show, and shall permit only the transitions shown in the state-transition diagram in Appendix B.

3.2 Search Vehicles and View Availability

3.2.1 Description and Priority

Any visitor may browse the catalogue of cars and two-wheelers, and any user may search for the vehicles that are genuinely free for a requested period at a chosen branch. Priority = High.

3.2.2 Stimulus/Response Sequences

Stimulus: User enters a pickup branch and a rental period and applies filters.
Response: System computes availability for the whole period and displays the matching vehicles with the estimated price.

Stimulus: User selects a vehicle.
Response: System displays the specification, tariff, deposit, terms, rating and date of last service.

Stimulus: No vehicle matches the criteria.
Response: System reports this and offers the nearest alternatives by branch or by date.

3.2.3 Functional Requirements

Search.Browse: The system shall permit an unauthenticated visitor to browse the catalogue and view specifications and published tariffs, but shall not permit a visitor to place a booking.

Search.Available: The system shall display a vehicle as available for a requested period only if its status is "Available," no Confirmed or Active booking overlaps the period, and no scheduled maintenance window overlaps it.

Search.Filter: The system shall permit the user to filter the results by vehicle type, category, brand and model, fuel type, transmission, seating capacity, price range and minimum rating, and to sort them by price or rating.

Search.Detail: The vehicle detail display shall show at least four photographs, the specification, the applicable tariff, the free-kilometre allowance, the excess-kilometre rate, the security deposit, the fuel and cancellation policies, the average customer rating and the date on which the vehicle was last serviced.

Search.Price: The system shall display for every vehicle in the result list the estimated price for the requested period, calculated per BR-6 and marked as an estimate that excludes usage-based charges.

3.3 Take Delivery of and Return a Vehicle

3.3.1 Description and Priority

Branch staff record the physical release of a vehicle to a customer and its return, including the odometer and fuel readings and a photographic condition report signed by the customer, and the system computes the charges arising from the rental. Priority = High.

3.3.2 Stimulus/Response Sequences

Stimulus: Customer presents the booking reference or pickup code at the branch.
Response: System displays the booking, the verified licence and the handover checklist.

Stimulus: Staff completes the handover checklist and the customer signs.
Response: System stores the signed condition report, sets the booking to "Active" and the vehicle to "On Rent."

Stimulus: Customer returns the vehicle and staff completes the return checklist.

Response: System computes the additional charges, displays an itemised settlement and settles it against the security deposit.

3.3.3 Functional Requirements

Handover.Refuse: The system shall prevent handover if the licence has expired, if the vehicle's insurance, pollution or fitness certificate is not valid for the whole rental period, or if the vehicle is flagged Unsafe, and shall state which condition was not met.

Handover.Checklist: The system shall require the odometer reading, the fuel or charge level, the accessories issued and at least four photographs covering all sides of the vehicle to be recorded at handover, with existing damage marked on a diagram of the vehicle.

Handover.Sign: The system shall present the condition report and the rental agreement to the customer for a digital signature, shall store the signed report as a document that cannot subsequently be altered, and shall e-mail a copy to the customer.

Handover.Activate: On completion of the handover the system shall set the booking status to "Active," set the vehicle status to "On Rent" and record the actual handover time.

Return.Checklist: The system shall require the odometer reading, the fuel or charge level, the accessories returned, the actual return time and at least four photographs to be recorded at return, and shall display the handover and return photographs together so that new damage can be identified.

Return.Odometer: The system shall reject a return odometer reading lower than the handover reading, and shall require a supervisor override with a recorded reason if the distance travelled exceeds the configured plausibility limit.

Return.Charges: The system shall compute and display as separate items the excess-kilometre charge per BR-11, the fuel shortfall per BR-12, the late return charge per BR-10, the cost of any accessory not returned, and any damage charge entered by the staff member with a justification.

Return.Settle: The system shall deduct the total additional charges from the security deposit, refund any balance per BR-14, and record a due against the customer if the charges exceed the deposit.

Return.Complete: On completion of the settlement the system shall set the booking status to "Completed," record the total distance travelled, and set the vehicle status to "Available" or, if the vehicle has been flagged, to "Under Maintenance."

Return.Flag: The system shall permit the staff member to flag the vehicle for maintenance during the return, stating the problem and its severity, and shall create a maintenance job automatically (Maint.Create).

3.4 Pay for a Rental and Receive Refunds

3.4.1 Description and Priority

The system collects the rental amount and the refundable security deposit, issues numbered invoices, settles usage-based charges at the end of the rental and issues refunds, maintaining a complete record of every transaction. Priority = High.

3.4.2 Stimulus/Response Sequences

Stimulus: Customer selects a payment method and submits payment.

Response: System requests authorisation from the payment gateway and, if it succeeds, confirms the booking and issues the invoice.

Stimulus: The payment gateway rejects the payment.

Response: System displays the reason, retains the hold for the remainder of its period, and lets the customer retry or choose another method.

Stimulus: A rental is completed with charges below the security deposit.

Response: System refunds the balance of the deposit and records the refund reference.

3.4.3 Functional Requirements

Pay.Method:	When the customer has confirmed the booking, the system shall ask the customer to select a payment method from card, on-line transfer, wallet or, for a booking to be collected at a branch, cash at the time of collection (see BR-9).
Pay.Authorize:	The system shall obtain authorisation from the payment gateway for the rental amount and the security deposit before confirming the booking.
Pay.Deposit:	The system shall hold the refundable security deposit defined in BR-3 separately from the rental amount, shall display it separately on every screen and document, and shall not treat it as revenue.
Pay.Confirm.OK:	If the authorisation is accepted, the system shall display a message confirming acceptance of the booking together with the booking reference and the transaction number.
Pay.Idempotent:	The system shall ensure that a repeated gateway notification, a duplicate submission or a refreshed browser page cannot result in the customer being charged twice for the same booking.
Pay.Invoice:	The system shall generate an invoice for every completed booking, bearing a unique sequential invoice number, the agency's tax registration details, an itemised list of all charges and the taxes applied. An issued invoice shall never be modified; a correction shall be made by a credit note that references the original invoice.
Pay.Ledger:	The system shall maintain an append-only ledger for every booking recording each charge, capture, refund, deposit hold, deposit release and adjustment with its amount, method, gateway reference, timestamp and the user who initiated it. A ledger entry shall never be deleted or overwritten.
Pay.Refund:	The system shall issue refunds to the original payment instrument, or, where the original payment was in cash, as a cash refund recorded by a named staff member and acknowledged by the customer.
Pay.Dues:	The system shall record any amount not recovered from the security deposit as a due against the customer, shall display it on the customer's home screen, and shall prevent further bookings until it is cleared (see BR-4). The system shall permit the customer to clear a due on-line at any time and shall lift the restriction immediately on payment.

3.5 View Rental History

3.5.1 Description and Priority

Each customer has a complete record of past, current and forthcoming rentals with the associated invoices and condition reports. Staff and Administrators have the corresponding operational view within their authority.
Priority = Medium.

3.5.2 Stimulus/Response Sequences

Stimulus:	Customer opens the rental history.
Response:	System displays all of that customer's bookings, most recent first.
Stimulus:	Customer selects a past booking.
Response:	System displays the full detail with links to the invoice, the condition reports and the payment record.

3.5.3 Functional Requirements

History.View:	The system shall permit a customer to view all of their own bookings, of every status, sorted with the most recent first, and to filter them by date range, status and vehicle type or search them by booking reference.
History.Download:	The system shall permit the customer to download or be sent by e-mail the invoice, the rental agreement and the condition reports for any of their bookings.

- History.Privacy: The system shall permit a customer to view only their own bookings, a staff member or Branch Manager only the bookings of their own branch, and an Administrator all bookings.
- History.Immutable: The system shall not permit any user to edit or delete a completed booking, its invoice or its condition reports. A correction shall be made only by an adjustment that references the original record.

3.6 Create, View and Close Vehicle Maintenance Records

3.6.1 Description and Priority

The system holds a complete service and repair history for every vehicle, raises preventive maintenance jobs automatically when a vehicle reaches its service interval, alerts the Administrator before statutory documents expire, and prevents any vehicle that is under maintenance or unfit for use from being booked. Priority = High.

3.6.2 Stimulus/Response Sequences

- Stimulus: Staff flags a vehicle at return, or a technician reports a fault.
 Response: System creates a maintenance job with status "Reported" and notifies the Administrator.
- Stimulus: A vehicle reaches its service interval in distance or elapsed time.
 Response: System raises a preventive maintenance job automatically and alerts the Administrator.
- Stimulus: Administrator schedules a job for a date range.
 Response: System blocks the vehicle's availability for that period and warns of any booking that conflicts.
- Stimulus: Technician records the work performed and closes the job.
 Response: System stores the record, computes the next service due, and releases the vehicle for booking.

3.6.3 Functional Requirements

- Maint.Create: The system shall permit a staff member, technician or Administrator to create a maintenance job specifying the vehicle, the job type as Preventive, Corrective, Inspection, Cleaning or Documentation, the reported problem, the severity and the reporting user.
- Maint.Auto.Km: The system shall create a Preventive job automatically when a vehicle's odometer reading reaches the distance-based service interval defined in BR-15, evaluated at every return.
- Maint.Auto.Time: The system shall create a Preventive job automatically when the time elapsed since the last completed service reaches the time-based interval defined in BR-15, evaluated daily.
- Maint.Schedule: The system shall permit an authorised user to schedule a job for a specific period, and shall block the vehicle's availability for that period so that it does not appear in search results.
- Maint.Conflict: If a proposed maintenance period overlaps a Confirmed booking, the system shall warn the user, list the affected bookings, and require explicit confirmation. On confirmation the system shall offer to notify the affected customers and to propose an alternative vehicle of the same category at the same tariff.
- Maint.Record: On completion the system shall require the service date, the odometer reading at service, the workshop or technician, a description of the work, each part replaced with its quantity and cost, the labour cost and the total cost to be recorded, and shall permit the workshop invoice to be attached.

Maint.Complete:	On completion of a job the system shall set the vehicle status back to “Available” unless it remains flagged, shall update the last service distance and date, and shall compute the next service due in distance and date per BR-15.
Maint.Status:	The system shall maintain every job in exactly one of the statuses Reported, Scheduled, In Progress, Completed or Cancelled, and shall permit only the transitions shown in Appendix B.
Maint.Unsafe:	The system shall permit an authorised user to flag a vehicle as Unsafe. A vehicle so flagged shall not appear in search results, shall not be bookable and shall not be handed over until the flag is cleared by an Administrator or authorised technician (see BR-16).
Maint.History:	The system shall display the complete chronological maintenance history of a vehicle to staff, technicians and Administrators, showing every job with its date, type, odometer reading, work performed, parts, cost and downtime.

3.7 Create, View, Modify and Delete Fleet, Tariffs and Users

3.7.1 Description and Priority

The Administrator maintains the vehicles, branches, tariffs, staff accounts and customer blacklist on which every other feature depends, with a complete audit trail of every change. Priority = High.

3.7.2 Stimulus/Response Sequences

Stimulus:	Administrator adds a vehicle to the fleet.
Response:	System validates the registration number, stores the master data, photographs and documents, and makes the vehicle bookable from its stated availability date.
Stimulus:	Administrator changes a tariff.
Response:	System records the new tariff with its effective date, leaves confirmed bookings and issued invoices unaffected, and logs the change.

3.7.3 Functional Requirements

Fleet.Add:	The system shall permit an Administrator to add a vehicle by recording the registration number, type, category, brand, model, year, colour, fuel type, transmission, seating capacity, chassis number, home branch, current odometer reading and availability date, together with at least four photographs and the expiry date of each statutory document.
Fleet.Status:	The system shall maintain every vehicle in exactly one of the statuses Available, On Rent, Under Maintenance, Unsafe or Retired, and shall display the current status of every vehicle on the fleet display.
Fleet.Retire:	The system shall permit an Administrator to retire a vehicle. A retired vehicle shall not appear in search results and shall not be bookable, but its rental, payment and maintenance history shall be retained and remain reportable. A vehicle with a Confirmed future booking shall not be retired until those bookings are cancelled or reassigned.
Fleet.Tariff:	The system shall permit an Administrator to define, for each vehicle category, the hourly, daily and weekly rate, the free-kilometre allowance, the excess-kilometre rate and the security deposit, each with an effective date.
Fleet.Tariff.History:	The system shall retain all historical tariffs. A Confirmed booking and an issued invoice shall always be priced at the tariff in force when the booking was confirmed, and shall never be altered by a later tariff change (see BR-13).
Fleet.Users:	The system shall permit an Administrator to create, edit, disable and re-enable Branch Staff, Maintenance Technician and Branch Manager accounts and to assign each to a branch. Disabling an account shall immediately end that user's sessions and refuse subsequent logins, and shall never alter the record of actions previously performed by that user.

Fleet.Audit: The system shall record in an audit log every creation, modification and deletion of vehicle, tariff, branch, user and blacklist data, with the user, the date and time and the previous and new values. The audit log shall be readable only by Administrators and shall not be editable by any user.

3.8 Notifications and Management Reports

3.8.1 Description and Priority

The system keeps customers and staff informed of everything that affects a booking, a payment or a vehicle, and provides the Administrator with the operational and financial information needed to run the fleet. Priority

= Medium, except the one-time password and booking confirmation messages, which are High because other features depend on them.

3.8.2 Stimulus/Response Sequences

Stimulus: A booking is confirmed, modified or cancelled.
Response: System sends an e-mail and an SMS message to the customer and updates the branch schedule.
Stimulus: A vehicle becomes overdue for return.
Response: System alerts the customer and the branch staff and states the charge that is accruing.
Stimulus: Administrator selects a report and a period.
Response: System computes and displays the report and offers it for export.

3.8.3 Functional Requirements

Notify.Booking: The system shall notify the customer by e-mail and SMS message on the confirmation, modification, extension, cancellation and no-show of a booking, stating the booking reference and the resulting status.
Notify.Remind: The system shall remind the customer 24 hours and 2 hours before the collection time, stating the branch, the time, the pickup code and the documents to bring, and 2 hours before the return time, stating the late return charge that will apply.
Notify.Payment: The system shall notify the customer of every successful payment, failed payment, settlement and change in the status of a refund.
Report.Dashboard: The system shall display to Administrators the number of vehicles in each status, the collections and returns due today, the bookings and revenue for the current day, week and month, the overdue returns and the open maintenance jobs.
Report.Utilisation: The system shall report vehicle utilisation, being the percentage of available hours during which a vehicle was on rent, by vehicle, category, branch and period, and shall identify the least utilised vehicles.
Report.Revenue: The system shall report revenue, refunds and net revenue by period, branch, vehicle category and payment method, distinguishing rental income from add-on income, and excluding security deposits.
Report.Access: An Administrator shall be able to report on all branches and a Branch Manager only on their own branch. No other user class shall have access to revenue or cost reports.

4. External Interface Requirements

4.1 User Interfaces

UI-1: The system shall present four interfaces, selected automatically according to the role of the logged-in user: the Customer interface, the Branch Staff interface, the Maintenance Technician interface and the Administrator interface.

- UI-2: Every screen shall display the application name, the identity of the logged-in user, a help link and a logout link in consistent positions.
- UI-3: The booking flow shall be presented as a progress-indicated sequence of steps, and the customer shall be able to move backwards through the steps without losing entered data.
- UI-4: Error messages shall be displayed next to the field that caused the error, shall state what is wrong and how to correct it, and shall never expose internal identifiers or database messages.
- UI-5: Every monetary amount shown to a user shall be itemised. A price shall never be displayed as a single total without the breakdown being available on the same screen.
- UI-6: The web pages shall permit complete navigation, vehicle selection and checkout using the keyboard alone, in addition to using mouse and keyboard combinations.
- UI-7: The screen displays shall conform to WCAG 2.1 Level AA, including a contrast ratio of at least 4.5:1 for body text, a text alternative for every image and correct behaviour at 200 percent zoom.
- UI-8: Any operation expected to take longer than two seconds shall display a progress indicator, and the control that started it shall be disabled to prevent duplicate submission.

4.2 Hardware Interfaces

- HI-1: The system shall use the camera of the customer's or staff member's device to capture driving licence images and vehicle condition photographs, compressed on the device before upload.
- HI-2: The system shall accept a vehicle identification tag scanned as a QR code, as an alternative to typing the registration number during handover and return.
- HI-3: The system shall support the capture of a customer's signature on a touch screen.

4.3 Software Interfaces

SI-1: Payment Gateway

- SI-1.1: The system shall request authorisation of the rental amount and the security deposit through a programmatic interface before a booking is confirmed.
- SI-1.2: The system shall request the capture of an authorised amount when the rental is completed and the final charges are known.
- SI-1.3: The system shall request the full or partial refund of a rental amount or a security deposit, and shall store the reference returned.
- SI-1.4: The system shall expose an endpoint to receive asynchronous payment and refund notifications, shall verify the signature of every notification, and shall process each notification idempotently so that a repeated notification does not create a duplicate transaction.

SI-2: Notification Service

- SI-2.1: The system shall send transactional e-mail through a mail service using versioned message templates.
- SI-2.2: The system shall send one-time passwords and alerts through an SMS gateway and shall record the delivery status returned for each message.
- SI-2.3: The system shall queue outbound messages so that a temporary failure of the notification service does not fail a booking transaction.

SI-3: Document Store

- SI-3.1: Licence images, condition photographs, signatures and generated documents shall be held in a document store, and only the object key shall be held in the database. Stored objects shall be retrieved only through signed links that expire within 15 minutes.

4.4 Communications Interfaces

- CI-1: All communication between the clients, the server and every external service shall use HTTPS with TLS 1.2 or higher and validated certificates.
- CI-2: The client and server shall communicate using REST over HTTP, exchanging JSON encoded as UTF-8.
- CI-3: The system shall send an e-mail message to the customer to confirm acceptance of a booking, the price and the collection instructions.
- CI-4: The system shall send an e-mail message to the customer to report any problem with the booking, the vehicle or the delivery after the booking is accepted.
- CI-5: The system shall send an SMS message containing the booking reference and pickup code within 60 seconds of confirmation, and a message containing a one-time password within 30 seconds of a verification request.
- CI-6: The system shall send push notifications to the mobile application for reminders and status changes where the application is installed and permission has been granted.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

- PE-1: The system shall accommodate 300 concurrent users during the peak usage window of 18:00 to 22:00 local time, with an estimated average session duration of 6 minutes.
- PE-2: A vehicle search, including the availability computation, shall return results within 3 seconds for 95 percent of queries and within 5 seconds for 99 percent of queries, over a 4 Mbps connection.
- PE-3: All other screens shall be fully rendered within 2 seconds over a broadband connection and within 5 seconds over a 4 Mbps mobile connection.
- PE-4: The availability and conflict check performed when a booking is confirmed shall complete within 1 second.
- PE-5: The system shall display the outcome of a payment attempt within 10 seconds of receiving the response from the gateway, and shall abandon the attempt and report a safe state if the gateway has not responded within 30 seconds.
- PE-6: Ninety-nine percent of database queries shall complete within 500 milliseconds with a data volume of 1,000 vehicles, 50,000 customers and 250,000 bookings.
- PE-7: A confirmation e-mail and SMS message shall be dispatched within 60 seconds of the event that caused it.

5.2 Safety Requirements

- SA-1: Before a two-wheeler booking is confirmed, the system shall display the safety notice, confirm that a helmet is included for each rider, and require the customer to acknowledge the notice. The acknowledgement shall be stored with the booking.

- SA-2: The system shall prevent the search, booking and handover of any vehicle flagged as Unsafe or overdue for a mandatory safety service. Only an Administrator or authorised technician may clear such a flag.
- SA-3: The system shall prevent the handover of a vehicle whose insurance policy, pollution certificate, fitness certificate or road tax is not valid for the whole booked period, and shall state which document is at fault.
- SA-4: The system shall record an emergency contact for every customer and shall display the 24-hour roadside assistance number on the booking confirmation, in the rental agreement and on the home screen of the mobile application while a rental is active.
- SA-5: The system shall not require a customer to interact with the application while a rental is active. Messages sent during an active rental shall be informational and shall be readable without any action being taken.

5.3 Security Requirements

- SE-1: All network transactions that involve financial information or personally identifiable information shall be encrypted in transit and the underlying documents encrypted at rest.
- SE-2: Passwords shall be stored only as salted hashes produced by bcrypt with a work factor of at least 12, and shall never be logged, e-mailed or displayed.
- SE-3: Users shall be required to log in for all operations except browsing the catalogue and viewing published tariffs.
- SE-4: The system shall enforce role-based access control on the server for every request. Hiding a control in the user interface shall never be the only mechanism preventing an unauthorised action.
- SE-5: The system shall permit a customer to view, modify or cancel only their own bookings, and to view only their own personal data, invoices and condition reports.
- SE-6: The system shall permit only staff members on the list of authorised Administrators to create or edit a tariff, and only Administrators or authorised technicians to close a maintenance job or clear an Unsafe flag.
- SE-7: The system shall verify possession of a mobile number by one-time password at registration and before any change to the registered mobile number or e-mail address.
- SE-8: After five consecutive failed login attempts the system shall lock the account for 15 minutes and notify the registered e-mail address. A failed login message shall not reveal whether the account exists.
- SE-9: A session shall expire after 30 minutes of inactivity, and re-authentication shall be required before a payment or a refund.
- SE-10: The system shall maintain a tamper-evident audit log of all authentication events, payment and refund transactions, tariff and fleet changes, licence verification decisions and administrative actions, retained for at least 12 months. No user shall be able to modify or delete an audit entry.
- SE-11: The system shall validate all input on the server and shall be free of the vulnerability classes listed in the OWASP Top Ten, in particular injection, broken access control, cross-site scripting and cross-site request forgery.

5.4 Software Quality Attributes

- Availability-1: The Vehicle Rental Management System shall be available to users 99.5 percent of the time between 06:00 and midnight local time and 98 percent of the time between midnight and 06:00. Planned maintenance shall take place between 02:00 and 05:00 and shall be announced at least 48 hours in advance.
- Reliability-1: No booking with a status of Confirmed shall ever overlap another Confirmed booking of the same vehicle. This shall be verified by a test that attempts 100 simultaneous confirmations of one vehicle, of which exactly

Robustness-1:	If the connection between the user and the system is broken before a booking is either confirmed or cancelled, the system shall enable the user to recover the incomplete booking for the remainder of the hold period.
Robustness-2:	The failure of an external service shall degrade the system gracefully. Every function that does not depend on that service shall remain usable, and the system shall state which function is temporarily unavailable.
Usability-1:	A customer who has not used the system before shall be able to complete a booking within 5 minutes without training. This shall be verified with at least 10 representative users, of whom at least 9 must succeed.
Maintainability-1:	The system shall be built as separate presentation, business logic and data access layers, so that a change to a business rule in Section 5.5 requires modification in one place only.
Maintainability-2:	Automated unit tests shall cover at least 70 percent of the business logic, and 100 percent of the pricing, availability and settlement logic.
Testability-1:	Every functional requirement in Section 3 shall be traceable to at least one test case. The system shall provide a test mode in which the payment, e-mail and SMS integrations use sandbox endpoints, so that the whole booking flow can be exercised without real money or messages.

5.5 Business Rules

BR-1:	A vehicle may be committed to only one confirmed booking at any instant, plus a turnaround buffer of 60 minutes between consecutive bookings for inspection and cleaning.
BR-2:	A customer's driving licence must be valid for the entire rental period, and the minimum age to rent is 18 years for a two-wheeler and 21 years for a car.
BR-3:	A customer may rent only those vehicle categories for which the licence is endorsed. A refundable security deposit of <CUR> 2,000 for a two-wheeler and <CUR> 5,000 for a car is collected for every rental.
BR-4:	A customer who is blacklisted, or who has an outstanding due greater than zero, may not create a new booking.
BR-5:	The minimum rental period is 4 hours and the maximum is 30 days, including any extension.
BR-6:	The rental charge is the applicable tariff for the period, plus add-on charges, plus usage-based charges, plus taxes, less any discount. A period exceeding 24 hours is billed in whole days and a shorter period in whole hours, a part of an hour being billed as a full hour after a grace period of 15 minutes.
BR-7:	Add-ons are charged at the published daily rate. One helmet per rider is provided free of charge with every two-wheeler rental.
BR-8:	The rental amount is refunded in full if a booking is cancelled at least 48 hours before collection, at 75 percent if cancelled between 12 and 48 hours before, and at 50 percent if cancelled less than 12 hours before. In the case of a no-show, 50 percent is retained. The security deposit is always refunded in full on cancellation.
BR-9:	Cash payment is accepted only for bookings collected at a branch, and must be recorded in the system by a named staff member when it is received.
BR-10:	A grace period of 15 minutes applies to the return time. Thereafter a late return is charged at the hourly rate, capped at one day's rate for each 24-hour period, and a delay exceeding 6 hours is billed as a full additional day.
BR-11:	Distance travelled beyond the free-kilometre allowance of the booking is charged at the excess-kilometre rate of the vehicle category.
BR-12:	The vehicle must be returned with the fuel or charge level at which it was issued. A shortfall is charged at the prevailing fuel or electricity price plus a service charge of 20 percent.

- BR-13: Only an Administrator may create or change a tariff. A tariff change takes effect from its effective date and never alters the price of a booking already confirmed or an invoice already issued.
- BR-14: The balance of the security deposit, after deduction of all charges due, is refunded to the original payment instrument within 7 working days of the rental being completed.
- BR-15: Preventive service is due every 5,000 km or 6 months for a car, and every 3,000 km or 4 months for a two-wheeler, whichever occurs first.
- BR-16: Only an Administrator or an authorised maintenance technician may close a maintenance job or clear an Unsafe flag, and the identity of that user is recorded with the action.

Appendix A: Data Dictionary and Data Model

booking	= booking reference + customer + vehicle + pickup branch + pickup date-time + return date-time + 0:m{add-on} + quoted amount + security deposit + booking status + pickup code
booking reference	= * a unique identifier assigned by the system to each confirmed booking; format VRMS-YYYYMMDD-NNNN *
booking status	= [draft pending payment confirmed active completed cancelled no-show] * see state-transition diagram in Appendix B *
condition report	= booking reference + report type + odometer reading + fuel level + 4:m{photograph} + 0:m{damage mark} + accessories + recorded by + recorded at + customer signature
customer	= customer id + customer name + customer e-mail + customer mobile + date of birth + address + (emergency contact) + licence + account status
invoice	= invoice number + booking reference + issue date + 1:m{charge line} + tax amount + total amount
licence	= licence number + issuing authority + 1:m{licence category} + issue date + expiry date + front image + back image + licence status
licence status	= [not submitted pending verification verified rejected expired]
maintenance job	= job id + vehicle + job type + reported problem + severity + job status + reported by + assigned to + scheduled from + scheduled to + (maintenance record)
job status	= [reported scheduled in progress completed cancelled]
odometer reading	= * the total distance recorded by the vehicle, in whole kilometres; may never decrease *
payment transaction	= transaction id + booking reference + transaction type + amount + payment method + gateway reference + transaction status + initiated by + timestamp
transaction status	= [initiated processing succeeded failed reversed]
vehicle	= registration number + vehicle type + vehicle category + brand + model + year + colour + fuel type + transmission + seating capacity + chassis number + home branch + odometer reading + vehicle status + 4:m{photograph} + 1:m{vehicle document}
vehicle status	= [available on rent under maintenance unsafe retired]

The principal entities and their relationships are shown in Figure 2. A Customer places many Bookings; a Vehicle is the subject of many Bookings over its life but of only one at any instant; each Booking produces one Invoice, two Condition Reports and many Payment Transactions; and each Vehicle accumulates many Maintenance Jobs, each consuming many Parts.

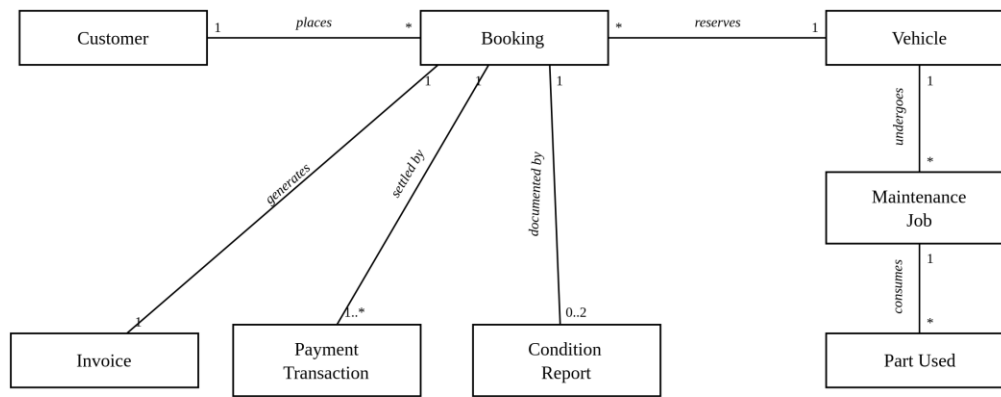


Figure 2. Partial data model for release 1.0 of the Vehicle Rental Management System.

Appendix B: Analysis Models

The status of a booking changes only through the transitions listed below. No other transition is permitted, and the system shall reject any request that would cause one.

Draft	→ Pending Payment, when the customer proceeds to payment and the system places a hold on the vehicle.
Pending Payment	→ Confirmed, when the payment is authorised, or a staff member records a cash payment.
Pending Payment	→ Draft, when the hold expires or the payment fails, releasing the vehicle.
Confirmed	→ Confirmed, when the customer modifies the period, branch or add-ons and the price is recalculated.
Confirmed	→ Active, when branch staff complete the handover and the customer signs the condition report.
Confirmed	→ Cancelled, when the customer cancels before collection; refund computed per BR-8.
Confirmed	→ No-show, when the customer has not collected the vehicle two hours after the pickup time.
Active	→ Active, when the customer extends the rental and pays the additional charge.
Active	→ Completed, when the vehicle is returned, the charges are settled and the deposit balance is refunded.

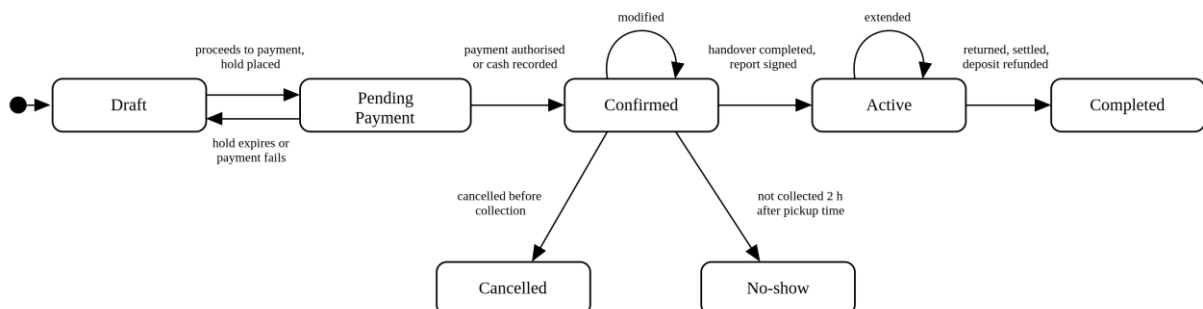


Figure 3. State-transition diagram for booking status.

The status of a vehicle and of a maintenance job change through the transitions below.

Available	→ On Rent, on completion of a handover; and → Under Maintenance, when a maintenance job is scheduled or started.
On Rent	→ Available, on completion of a return; or → Under Maintenance, if staff flagged the vehicle during the return.

Under Maintenance → Available, when the maintenance job is completed and the vehicle is released.

Any status → Unsafe, when an authorised user flags the vehicle; → Available again only when an Administrator or authorised technician clears the flag; → Retired, when the Administrator retires the vehicle, which requires that no future booking exists.

Reported → Scheduled → In Progress → Completed for a maintenance job; a job may move to Cancelled from Reported or Scheduled only.

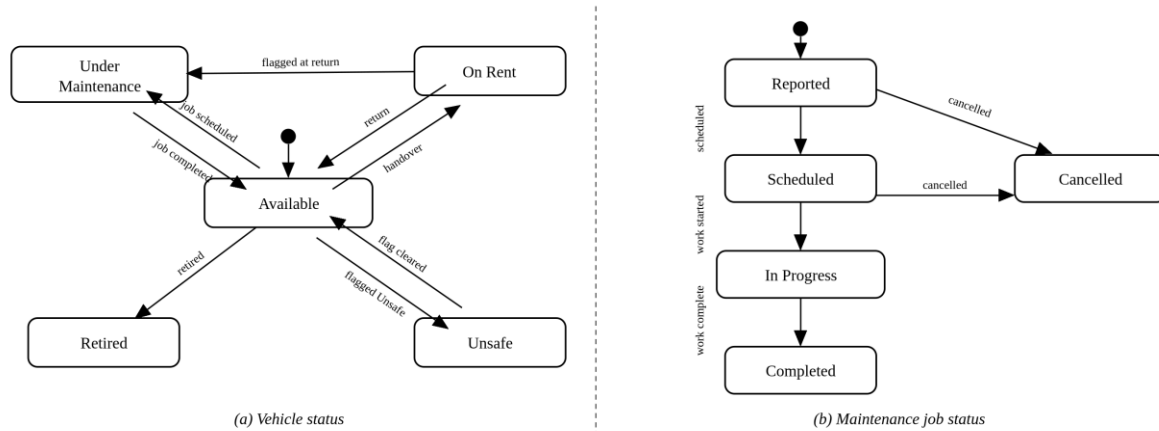


Figure 4. State-transition diagrams for vehicle status and maintenance job status.